

lowest RGB tone readings (4.31, 3.11, and 4.35 respectively), while the Acer ScanPrisa 640U registered the best combined RGB value, with the individual readings

Scanning Transparencies

If you want to scan transparencies, slides or negatives, you have to place the original between the light source and the sensor so that light passes through it before it reaches the sensor. But most consumer-level transparency scanners place both the light source and the sensor on one side. Due to this, the sensor can actually pick up light as it reflects off the original. In such cases, you will need a transparency adapter (TPA), which is nothing but a backlight to replace the cover of the scanner. Not all scanners support TPAs, and the adapter costs between Rs 4,500 and Rs 13,500. Some high-end scanners come packed with the TPA.

An alternative method is to have a slide-out tray so that there is no glass surface between the original and the sensor. This reduces what are called Newton Rings. The downside is that these scanners cost around Rs 36,000.

Yet another option is to go in for a slide or transparency (4x5) scanner. These scanners focus on a small area and provide a high optical resolution for enlarging small negatives or slides. But good slide scanners are expensive, costing between Rs 40,000 and Rs 90,000.

The feature that you need to consider for scanning negatives or slides is optical density (OD). The higher the OD, the better the scanner. The OD is usually associated with the colour depth. For instance, the OD of a 24-bit scanner is generally less than that of a 36-bit scanner. But scanners with the same colour depth can have different optical densities.

Most scanners today have an OD of about 2.9, which is sufficient for scanning normal pictures. Slides and transparencies scan better with an OD of at least 3.2, and negatives need an OD of 3.4. An easy way of differentiating between a good scanner and a consumer-level device is to look at the packaging—a consumer-level scanner will never have the OD printed on its specs sheet.

being 11.65, 6.48 and 6.3, respectively. This implies that the ScanPrisa 640U's CCD (charged couple device) is of very high quality since it is wholly responsible for the detection of variations in colour tones.

In the last test in this series, we evaluated the tone detection in the shadow and bright areas of the IT8 card. We selected the area from block 1 to block 4 (F1-F4) for detecting variations in the shadows, and block 9 to block 12 (F9-F12) to test the detection of tones in the bright areas. Two blocks were selected, for instance, F1



Microtek ScanMaker 3600

and F2. The maximum area was selected and the standard deviation of the individual RGB values was noted down for both the shadow and bright regions of the IT8 card. Many of the scanners had problems in detecting the green tone and scored less than 2. The scanner that did well in this test was the HP ScanJet 6350C. It logged impressive results in the detection of red tones in the shadow region, but performed poorly in detecting the green tone, as did all the other scanners that we tested.

The scanners that gave consistent readings in all the three tests were the HP ScanJet 5300C and the Microtek ScanMaker

3600. The Umax AstraNET iA101 turned in the poorest performance—the standard deviation score for red was a dismal 0.47. The Acer ScanPrisa 640U also recorded poor readings in these tests.

Resolution test

This test evaluates the ability of the optics used in the scanner to detect the smallest possible details on the target document. We used a resolution card containing five blocks, each of which is made up of very fine lines that are created at a particular resolution. The first block was of 120 ppi (points per inch) while the fifth block was of 2400 ppi. The scan was done both in full colour and gray scale modes, and at the maximum optical resolution and bit-depth. It was then converted to 8-bit mode in order to

use Magic Wand, which doesn't work on the directly scanned image. The tolerance was kept at 8 as opposed to the default value of 32. The individual scans were checked to ascertain the scanner's ability to pick up a particular resolution. All the scanners passed the third block effortlessly in gray scale mode, but many of them failed to detect even a single line in the fourth block (1200 ppi). The Microtek ScanMaker 3600 and the Umax Astra 3400 failed to detect even the second block in full colour mode.

The winner in this test was the Umax Astra 4000U, which detected all the individual lines in the fifth block. This scanner returned above-average results in the other tests but could not make it to the top three. The only other scanner that matched this performance was the Epson Perfection 1240U Photo. Although it could not detect the fifth block in full colour mode, it managed to do so in gray scale mode.

